In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

FULL TEST – V

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 16) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 or 5 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

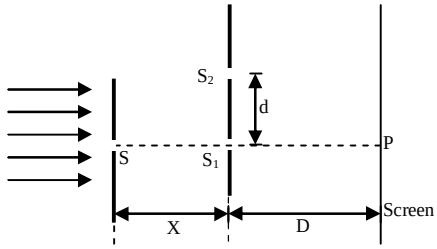
SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

1. A heavy nucleus at rest breaks into two fragments which fly off with velocities in the ratio of 8 : 1. The ratio of the radii of the fragments (assumed spherical) is
   (A) 1 : 2 (B) 1 : 4
   (C) 4 : 1 (D) 2 : 1

2. Monochromatic light of wavelength λ emerging from slit S illuminates slits S_1 and S_2 which are placed with respect to S as shown in Fig. 27.5. The distances x and D are large compared to the separation d between the slits. If x = D/2, the minimum value of d so that there is a dark fringe at the centre P of the screen is



   (A) $\sqrt{\frac{\lambda D}{3}}$ (B) $\sqrt{\frac{2\lambda D}{3}}$
   (C) $\sqrt{\lambda D}$ (D) $2\sqrt{\frac{\lambda D}{3}}$

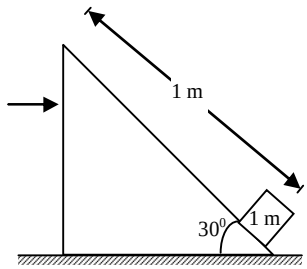
3. A particle free to move along the x-axis has potential energy given by
   $U(x) = k[1 - \exp(-x^2)]$ for $-\infty \le x \le +\infty$
   where k is a constant of appropriate dimension.
   Then
   (A) at points away from the origin, the particle is in unstable equilibrium
   (B) for any finite nonzero value of x, there is a force directed away from the origin
   (C) if its total mechanical energy is k/2, it has its minimum kinetic energy at the origin
   (D) for small displacements from x = 0, the motion is simple harmonic

Space for rough work

4. Three rods of the same length are arranged to form an equilateral triangle. Two rods are made of the same material of coefficient of linear expansion α_1 and the third rod which forms the base of the triangle has coefficient of expansion α_2. The altitude of the triangle will remain the same at all temperatures if the α_1 / α_2 is nearly

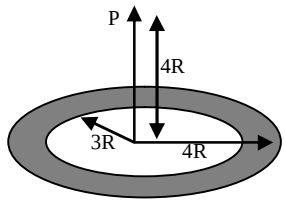
(A) 1 (B) 1/2
(C) 1/4 (D) 4

5. In the figure, the wedge is pushed with an acceleration of $10\sqrt{3}$ m / s^1. It is seen that the block starts climbing upon the smooth inclined face of wedge. What will be the time taken by the block to reach the top?



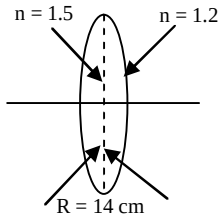
(A) $\frac{2}{\sqrt{5}}$s (B) $\frac{1}{\sqrt{5}}$s
(C) $\sqrt{5}$ s (D) $\sqrt{\frac{5}{2}}$ s

6. A thin uniform annular disc (see figure) of mass M has outer radius 4R and inner radius 3R. The work required to take a unit mass from point P on its axis to infinity is



(A) $\frac{2GM}{7R}\left(4\sqrt{2}-5\right)$ (B) $-\frac{2GM}{7R}\left(4\sqrt{2}-5\right)$
(C) $\frac{GM}{4R}$ (D) $\frac{2GM}{5R}\left(\sqrt{2}-1\right)$

7. A bi-convex lens is formed with two thin planoconvex lenses as shown in the figure. Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surface are of the same radius of curvature R = 14 cm, For this Bi-convex lens, for an object distance of 40 cm, the image distance will be

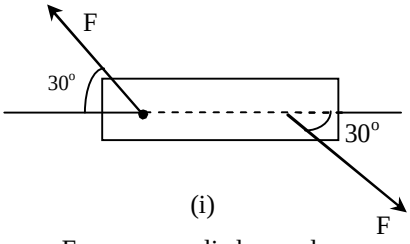


(A) -280.0 cm (B) 40.0 cm
(C) 21.5 cm (D) 13.3 cm

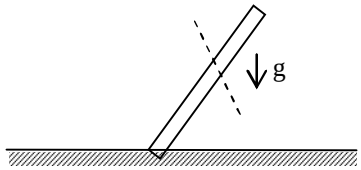
Space for rough work

8. The half-life of ^{131}I is 8 days. Given a sample of ^{131}I at time t = 0, we can conclude that
(A) no nucleus will decay before t = 4 days
(B) no nucleus will decay before t = 8 days
(C) all neclei will decay before t = 16 days
(D) a given nucleus may decay any time after t = 0

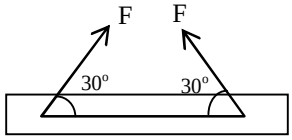
9. Which of cases have a non zero net torque activity on the rod about its centre?



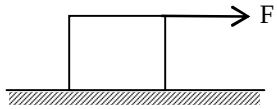
(i)
Force are applied on rod



(ii
Rod sliding on smooth ground



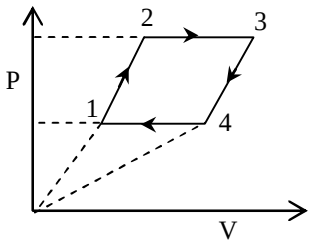
(iii)



(iv)
(Cube is pulled on smooth surface and moving with constant acceleration)

(A) In case (i), (ii)
(B) In case (ii), (iv), (i)
(C) In case (i) & (iii)
(D) In all case

10. Three moles of an ideal monatomic gas performs a cycle shown in figure then temperature of the gas is different states asked as 1, 2, 3 and 4 in figure are 400, 700, 2500 & 1100 K respectively
(A) value of $= P_2 / P_1 = V_4 / V_3$
(B) value of $P_2 / P_1 = V_3 / V_4$
(C) work done = 14 J
(D) work done $= 18.247$J

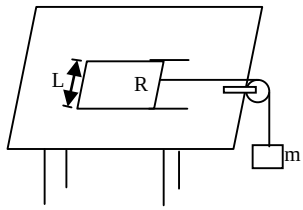


Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Questions 11 & 12



A pair of parallel horizontal conducting rails of negligible resistance shorted at one end is fixed on a table. The distance between the rails is L. A conducting massless road of resistance R can slide on the rails without friction. The rod is tied to a massless string which passes over a pulley fixed to the edge of the table. A mass m, fixed to the other end of the string, hang vertically. A constant magnetic field B exists perpendicular to the table. The system is released from rest.

11. The acceleration the mass m moving in the downward direction is

(A) g (B) $\frac{B^2L^2v}{MR}$

(C) $\left(g-\frac{B^2L^2v}{MR}\right)$ (D) $\left(g+\frac{B^2L^2v}{mR}\right)$

12. The acceleration of mass m when the velocity of the rod is half the terminal velocity is

(A) g (B) g/2

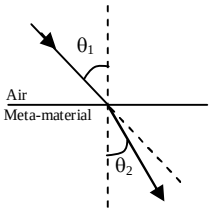
(C) g/3 (D) g/4

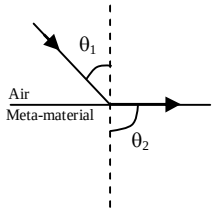
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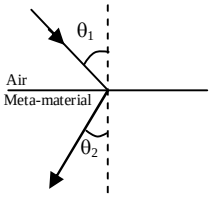
Paragraph for Questions 13 & 14

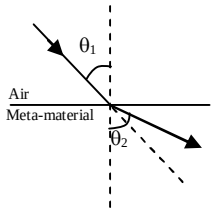
Most materials have the refractive index, n > 1. So, when a light ray from air enters naturally occurring material, then by snell's law, $\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$, it is under stood that the refracted ray bends towards the normal. But is never emerges on the same side of the normal as the incident ray. According to electromagnetism, the refractive index of the medium is given by the relation, $n = \left(\frac{C}{v}\right) = \pm\sqrt{\varepsilon_r \mu_r}$ where ε is the speed of electromagnetic waves in vacuum, v its speed in the medium, ε_r, and μ, are relative permittivity and permeability of medium respectively. In normal materials, ε_r and μ_r, are positive, implying positive n for the medium. When both ε_r and μ_r, are negative, one must choose the negative root of n. Such negative refractive index materials can now be artificially prepared and are called meta-materials. They exhibit significantly different optical behavior, without violating any physical laws. Since n is negative, it results in a change in the direction of propagation of the refracted light. However, similar to normal materials, the frequency of light remains unchanged upon refraction even in meta-materials.

13. For light incident from air on a meta-material, the appropriate ray diagram is

(A) 

(B) 

(C) 

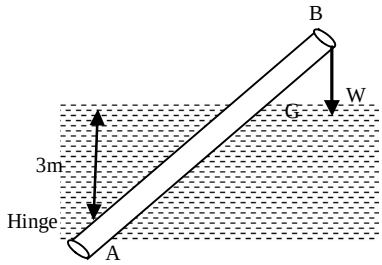
(D) 

Space for rough work

14. Choose the correct statement:
(A) The speed of light in the meta-material is $v = c|n|$
(B) The speed of light in the meta-meterial is $v = \frac{C}{n}$
(C) The speed of light in the meta-material is v = c
(D) The wavelength of the light in the meta-material (λ_m) is given $\lambda_m = \lambda_{air}|n|$. Where λ_{air} is wavelength of the light in air.

Paragraph for Questions 15 & 16

A rod of length 6 m has a mass 12 kg. it is hinged at on end A at a distance of 3 m below water surface. The specific gravity of the material of rod is 0.5.



15. What weight must be attached to the other end B so that 5 m of the rod is immersed in water?
(A) 7 kgf
(B) $\frac{7}{3}$ kgf
(C) $\frac{7}{5}$ kgf
(D) $\frac{7}{2}$ kgf

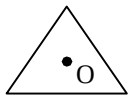
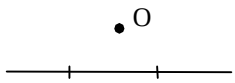
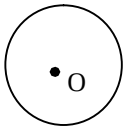
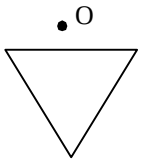
16. Find the magnitude and direction of the force exerted by the hinge on the rod:
(A) $\frac{17}{3}$ kgf in the downward direction
(B) 8 kgf in the downward direction
(C) 4 kgf in the downward direction
(D) 5 kgf in the downward direction

Space for rough work

(Match List Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which only **ONE** is correct.

17. Three wires of length l and linear charge density λ each is shaped into an equilateral triangle with point O as centroid of this triangle. Now the wires are arranged in different shapes with point O fixed as defined earlier. Math List – I having different shapes with List – II potential at point O:

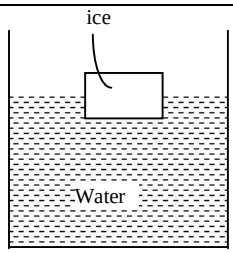
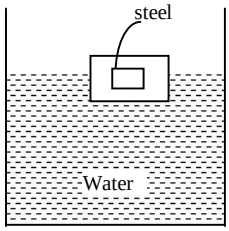
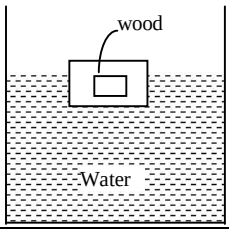
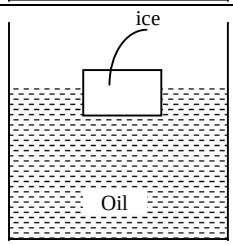
List - I		**List - II**	
(P)		(1)	$\frac{\lambda}{\pi\varepsilon_0}\ln\left(2+\sqrt{3}\right)$
(Q)		(2)	$\frac{\lambda}{2\pi\varepsilon_0}\ln\left(2\sqrt{7}+3\sqrt{3}\right)$
(R)		(3)	$\frac{3\lambda}{2\pi\varepsilon_0}\ln\left(2+\sqrt{3}\right)$
(S)		(4)	$\frac{\pi\lambda}{2\pi\varepsilon_0}$

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	1	4	2	3
(C)	3	2	4	1
(D)	3	4	2	1

Space for rough work

18. When ice melts completely, level of liquid in which ice is submerged:

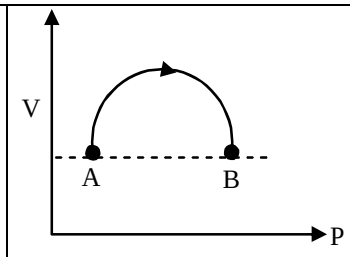
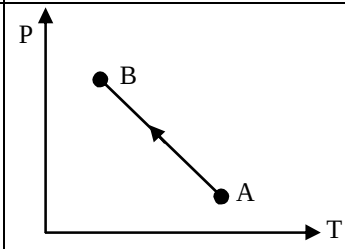
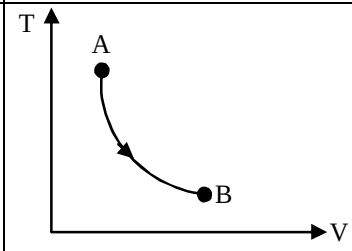
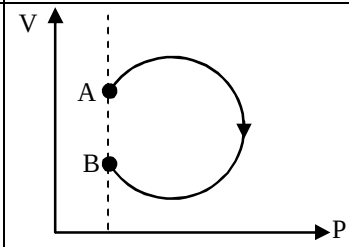
List – I		List – II	
(P)		(1)	Increases
(Q)		(2)	Decreases
(R)		(3)	Remains same
(S)	 Density of oil is greater than density of ice	(4)	May increases or decreases

Codes:

	P	Q	R	S
(A)	1	2	4	4
(B)	3	2	4	4
(C)	3	1	4	1
(D)	3	2	3	1

Space for rough work

19. A sample of gas goes from state A to state B in four different manners, as shown by the graphs. Let W be the work done by the gas and ΔU be change in internal energy along the path AB. Correctly match the graphs with the statements provided.

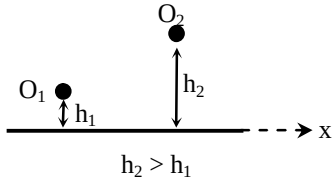
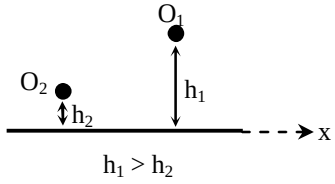
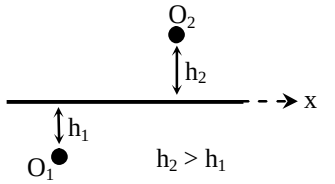
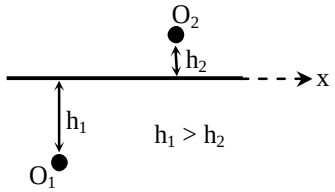
List – I		List – II	
(P)		(1)	Both W and ΔU are positive
(Q)		(2)	Both W and ΔU are negative
(R)		(3)	W is positive whereas ΔU is negative
(S)		(4)	W is negative whereas ΔU is positive
		(5)	Specific heat of gas must be positive

Codes:

	P	Q	R	S
(A)	4	2,5	3	2,5
(B)	4	2,5	1	2,5
(C)	4,5	2,5	3	4,5
(D)	5	3,5	3,5	3

Space for rough work

20. Column shows O_1 object and O_2 image. The optical system responsible can be a spherical mirror, plane mirror or a thin lens. In the case of the lens and spherical mirror, straight line shows the principal axis. Match List-I and List-II:

List – I		**List – II**	
(P)		(1)	Concave mirror between O_1 and O_2
(Q)		(2)	Diverging lens between O_1 and O_2
(R)		(3)	Convex mirror between O_1 and O_2
(S)		(4)	Converging lens between O_1 and O_2
		(5)	Inclined plane mirror somewhere between O_1 and O_2

Codes:

	P	Q	R	S
(A)	1,2,5	3,5	4,5	4,5
(B)	1,5	2,5	1,2,5	1,4,5
(C)	1,2,5	3,4,5	4,5	3,4,5
(D)	1,5	3,5	4,5	4,5

Space for rough work

Chemistry

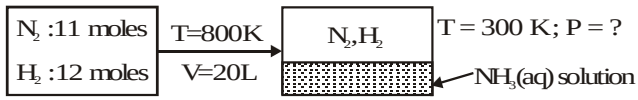
PART – II

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

1. 11 moles N_2 and 12 moles of H_2 mixture reacted in 20 litre vessel at 800 K. After equilibrium was reached, 6 mole of H_2 was present. 3.58 litre of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300 K. What is the final pressure of gaseous mixture? Neglect vapour pressure of liquid solution. Assume (i) all NH_3 dissolved in water (ii) no change in volume of liquid (iii) no reaction of N_2 and H_2 at 300 K :



Initial condition

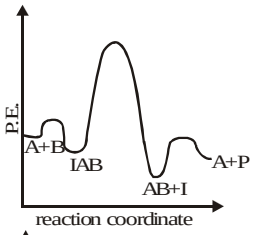
(A) 18.47 atm
(B) 60 atm
(C) 22.5 atm
(D) 45 atm

2. The following mechanism has been proposed for the exothermic catalyzed complex reaction:

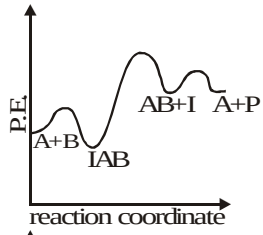
$$A + B \underset{}{\overset{\text{fast}}{\rightleftharpoons}} I\,AB \overset{k_1}{\rightleftharpoons} AB + I \overset{k_2}{\rightleftharpoons} P + A$$

If k_1 is much smaller than k_2, the most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

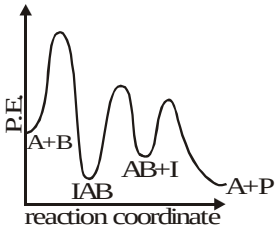
(A)



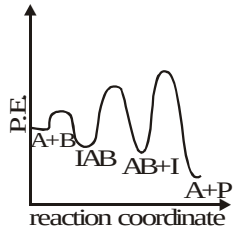
(B)



(C)



(D)

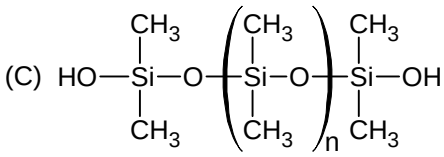


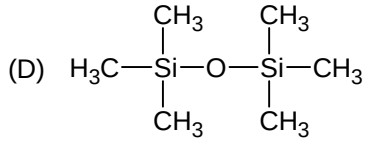
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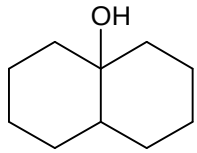
3. The final product of complete hydrolysis of $(CH_3)_2SiCl_2$ followed by polymerization forms

(A) $(CH_3)_2SiOSi(CH_3)_2$

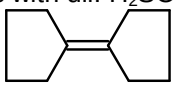
(B) $(CH_3)_2Si(OH)_2$

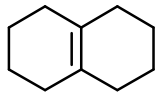
(C) 

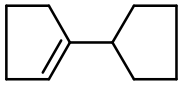
(D) 

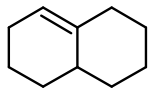
4. Which of the following alkene is obtained as major product when the alcohol 

reacts with dil. H_2SO_4?

(A) 

(B) 

(C) 

(D) 

5. At what $\frac{[Br^-]}{\sqrt{[CO_3^{2-}]}}$ does the following cell have its reaction at equilibrium?

Ag(s) | Ag_2CO_3(s) | Na_2CO_3 (aq) || KBr(aq) | AgBr(s) | Ag(s)

$K_{SP} = 8 \times 10^{-12}$ for Ag_2CO_3 and $K_{SP} = 4 \times 10^{-13}$ for AgBr

(A) $\sqrt{1} \times 10^{-7}$

(B) $\sqrt{2} \times 10^{-7}$

(C) $\sqrt{3} \times 10^{-7}$

(D) $\sqrt{4} \times 10^{-7}$

6. Which of the following species do not have $O-O$ bond length longer than that in O_2 molecule?

(A) $H_2S_2O_8$

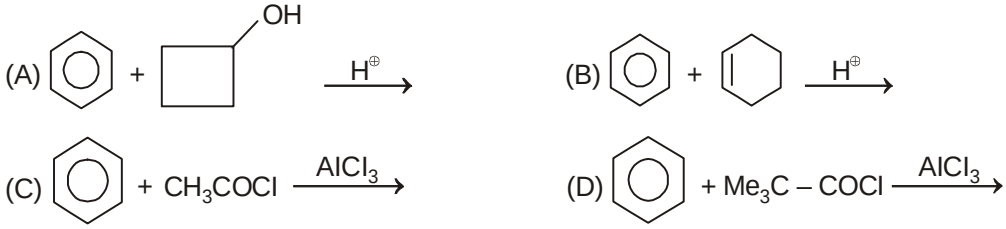
(B) H_2SO_5

(C) H_2O_2

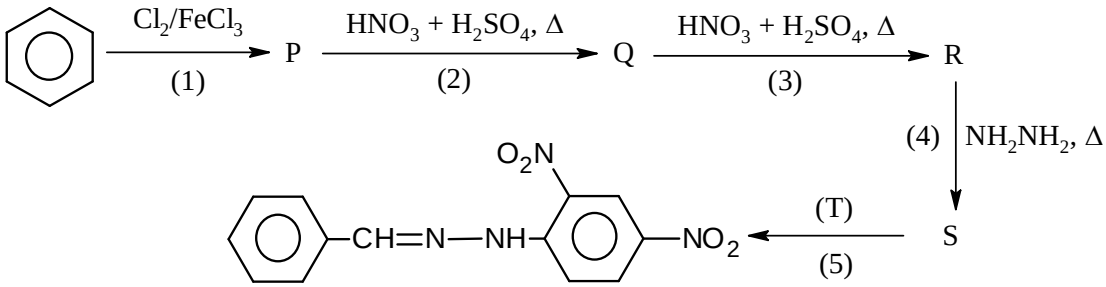
(D) $O_2[BF_4]$

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7. Which of the following reactions will not give alkylation product?

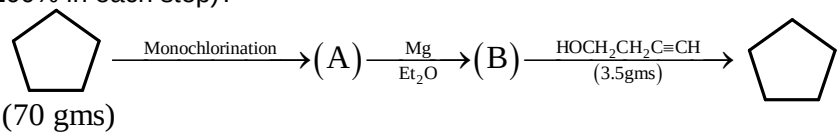


8. The incorrect statement about the following reaction sequence is



(A) 'T' gives an aldol condensation reaction on heating with NaOH solution
(B) Compound R is more reactive towards N_2H_4 than compound Q.
(C) compound $Q \xrightarrow[(ii)\,NaNO_2-HCl,\ 0-5^0\,C]{(i)\,Sn/HCl} X \xrightarrow{H_3PO_2} P$; P is PhCl
(D) In step 5 Acetaldehyde is more reactive than T

9. How many gms of cyclopentane will be formed in the reaction given below (Consider the yield be 100% in each step)?



(70 gms)

(A) 5 (B) 6
(C) 7 (D) 8

10. The number of vacant hybrid orbitals which participate in the formation of 3-centre 2 electron bonds i.e., banana bonds in diborane structure are:
(A) 0 (B) 1
(C) 2 (D) 3

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Questions 11 & 12

In the manufacture of $Na_2CO_3(s)$ by Solvay process, $NaHCO_3(s)$ is decomposed by heating:

$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$

$K_p = 0.23$ at 100°C $\Delta H^o = 136$ kJ

11. If a sample of $NaHCO_3$ (s) is brought to a temperature of 100°C in a closed container total gas pressure at equilibrium is:
(A) 0.96 atm (B) 0.23 atm
(C) 0.48 atm (D) 0.46 atm

12. A mixture of 1.00 mol each of $NaHCO_3(s)$ and $Na_2CO_3(s)$ is introduced into a 2.5 L flask in which $P_{CO_2} = 2.10$ atm and $P_{H_2O} = 0.94$ atm.When equilibrium is established at 100°C, then partial pressure of
(A) $CO_2(g)$ and $H_2O(g)$ will be greater than their initial pressure
(B) $CO_2(g)$ and $H_2O(g)$ will be less than their initial pressure
(C) $CO_2(g)$ will be larger and that of $H_2O(g)$ will be less than their initial pressure
(D) $H_2O(g)$ will be larger and that of $CO_2(g)$ will be less than their initial pressure

Paragraph for Questions 13 & 14

The empirical formula of a cobalt (III) complex is $Co(NH_3)_X Cl_3$. The coordination no.of cobalt in the above complex is six. Say that all complexes ionize.

13. K_b of water 2K kg $mole^{-1}$, when one mole of the above complex is added to 1kg water the rise in B.Pt is 6K. Then the value of x is
(A) 4 (B) 3
(C) 5 (D) 6

14. One mole of the above complex gave 143.5g white ppt with excess $AgNO_3$ solution. Then x is
(A) 4 (B) 5
(C) 6 (D) 3

Space for rough work

Paragraph for Questions 15 & 16

A white crystalline solid (A) on dehydration gave a poisonous pseudo halogen gas (B). Compound (A) when filed with alkali gave a colourless gas (C), which forms white fumes with HCl. The fumes condenses to white solid with HCl. The fumes condenses to white solid (D) on cooling. (B) dissolves in KOH solution to gave two compounds (pseudo halides) (E) and (F). Compound (E) gave white ppt. (G) with $AgNO_3$ solution, but ppt (G) dissolves on addition of excess (E). When (F) is heated together with (D) a well known fertilizer (H) is obtained.

15. As per the above passage (A) is

(A) $(NH_4)_2C_2O_4$ (B) $(NH_4)_2CO_3$

(C) NH_4CN (D) NH_4BrO_3

16. Which of the following dehydration agent can not be used for the conversion of (A) to (B)?

(A) P_2O_5 (B) conc.H_2SO_4

(C) CaO (D) (B) and (C) both

(Match List Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which only **ONE** is correct.

17. Match the following:

List – I		**List – II**	
(P)	Isothermal process	(1)	$\Delta E = 0$
(Q)	Reversible adiabatic process	(2)	$\Delta H = 0$
(R)	Cyclic process	(3)	$\Delta S = 0$
(S)	Isochoric process	(4)	$w = 0$
		(5)	$\Delta G = 0$

Codes:

	P	Q	R	S
(A)	1, 2	3	1,2,3,5	4
(B)	2	4	4, 5	1
(C)	1	2	3	5
(D)	4	5	2	3, 4

Space for rough work

18. Match the following:

List – I		List – II	
(P)	O_2	(1)	Paramagnetic
(Q)	B_2	(2)	Día magnetic
(R)	C_2	(3)	Bond order : 2
(S)	N_2	(4)	Last e^- goes in anti-bonding molecular orbital
		(5)	Last e^- goes in bonding molecular orbital

Codes:

	P	Q	R	S
(A)	2,4	3,4	4	5
(B)	1,3,4	1,5	2,3,5	2,5
(C)	2,5	2,3,4	4	3,5
(D)	4	3,5	4,5	1,3,4

19. Match the following:

List – I		List – II	
(P)	Addition of HCl can make the buffer solution	(1)	CH_3COONa
(Q)	Addition of NaOH can make the buffer solution	(2)	NH_3
(R)	Increase solubility when added to a saturated solution of $CuCO_3$	(3)	NH_4Cl
(S)	Form buffer on mixing with solution of their conjugate acid base	(4)	$NaHCO_3$
		(5)	HCOONa

Codes:

	P	Q	R	S
(A)	1, 2, 4, 5	3, 4	2	1, 2, 3, 4, 5
(B)	2	3	4	5
(C)	5, 1	2, 4	3, 5	2, 4
(D)	5, 1	3, 5	4, 5	5, 6

20. Match the following:

List – I		List – II	
(P)	$NO_{(g)} + NO_{3(g)} \rightleftharpoons 2NO_{2(g)}$	(1)	$K_p > K_c$
(Q)	$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$	(2)	$K_p = K_c(RT)^2$
(R)	$2O_{3(g)} \rightleftharpoons 3O_{2(g)}$	(3)	$K_p = K_c$
(S)	$BrF_{5(g)} \rightleftharpoons \frac{1}{2}Br_{2(g)} + \frac{5}{2}F_{2(g)}$ note : T > 300°C	(4)	$K_p < K_c$
		(5)	$K_p = K_c(RT)^1$

Codes:

	P	Q	R	S
(A)	3	4	1, 5	1,2
(B)	4	5, 3	5, 1	1
(C)	5,1	2, 5	4, 2	3
(D)	1	2	3	4

Space for rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

1. A triangle ABC has incentre I. Points X, Y are located on the line segment AB, AC respectively so that BX. AB = IB^2 and CY. AC = IC^2. Given that X, I, Y are collinear. The possible values of measure of angle A is

(A) 120° (B) 60°

(C) 30° (D) cannot be determined

2. An equilateral triangle of side length 2 units is inscribed in a circle. The length of a chord of this circle which passes through the midpoints of two sides of this triangle is

(A) $\frac{\sqrt{5}}{2}$ (B) $\sqrt{5}$

(C) $\frac{2}{3}$ (D) $\frac{\sqrt{2}}{3}$

3. $N = 7^{\log_{49} 900}$, $A = 2^{\log_2 4} + 3^{\log_2 4} + 4^{\log_2 2} - 4^{\log_2 3}$, $D = (\log_5 49)(\log_7 125)$.

The value of $\log_{\left(A-\frac{N}{10}\right)} |N + A + D + 6| - \log_5 2$ is equal to

(A) 1 (B) 2

(C) 3 (D) 4

4. Let $a_1, a_2, \ldots\ldots$ be real constants and $y(x) = \cos(a_1 + x) + \frac{1}{2}\cos(a_2 + x) + \frac{1}{2^2}\cos(a_3 + x) + \ldots\ldots + \frac{1}{2^{n-1}}\cos(a_n + x)$. If $x_1 x_2$ are real & $y(x_1) = y(x_2) = 0$, then $x_2 - x_1 =$

(A) $n\pi, n \in I$ (B) $\frac{n\pi}{2}, n \in I$

(C) $\frac{n\pi}{3}, n \in I$ (D) $\frac{n\pi}{4}, n \in I$

Space for rough work

5. Let a_n & b_n are positive integers and $a_n + \sqrt{2}b_n = \left(2+\sqrt{2}\right)^n$ then $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)$ =

(A) $\sqrt{2}$ (B) 2

(C) $e^{\sqrt{2}}$ (D) e^2

6. For any positive integer n, $a_n = \sum_{k=1}^{n}\tan^2\frac{k\pi}{2n+1}$ & $b_n = \prod_{k=1}^{n}\tan^2\frac{k\pi}{2n+1}$, then the value of $\frac{a_n}{b_n}$ is equal to

(A) $\tan^2\frac{n\pi}{n^2+1}$ (B) $n\pi$

(C) n (D) $\frac{n\pi}{n^2+1}$

7. An equation $a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_{99}x^{99} + x^{100} = 0$ has roots ${}^{100}C_r \sin(r\theta)$, where r = 1, ...100 , then the value of a_{99} is

(A) $\left(\cos\frac{\theta}{2}\right)^{100}\sin 50\theta$ (B) $2^{100}\left(\cos\frac{\theta}{2}\right)^{100}$

(C) $2^{100}\sin 50\theta$ (D) $2^{100}\left(\cos\frac{\theta}{2}\right)^{100}\sin 50\theta$

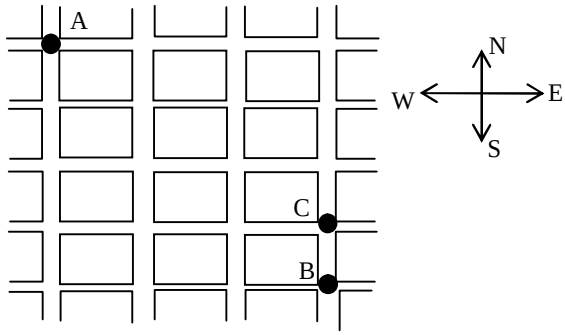
8. In ΔABC, there is a interior point P such that $\angle PAB = 10^o, \angle PBA = 20^o$, $\angle PCA = 30^o$ & $\angle PAC = 40^o$, then $\angle ACB$ =

(A) 50° (B) 40°

(C) 20° (D) 10°

Space for rough work

9. The adjoining figure is a map of a part of a city. The small rectangles are blocks & spaces in between are streets. Each morning a student walks from intersection A to intersection B, always walking along streets shown, always going east or south. For variety, at each intersection where he has a choice, he chooses with probability ½ (independent of all other choices) whether to go east or south. The probability that, on any given morning, he walks through intersection C is



(A) $\frac{11}{32}$ (B) $\frac{1}{2}$

(C) $\frac{4}{7}$ (D) $\frac{21}{32}$

10. If $f(0) = 0$ and f is differentiable at x = 0 and k is positive integer then $\lim_{x\to\infty} \frac{1}{x}\left[f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{3}\right) + ... + f\left(\frac{x}{k}\right)\right]$ is equal to

(A) $k.\, f'(0)$ (B) $\left(\sum_{r=1}^{k} \frac{1}{r}\right) f'(0)$

(C) $\sum_{r=1}^{k} \frac{1}{r}$ (D) does not exist

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Questions 11 & 12

A function of satisfies the equation $f(x).f'(-x) = f(-x)f'(x)$ for all x and f(0) = 3.

11. The value of $f(x).f(-x)$ for all *x* is

(A) 1 (B) 4
(C) 9 (D) 16

12. $\int_{-51}^{51} \frac{dx}{3+f(x)}$ is equal to

(A) 17 (B) 34
(C) 102 (D) 0

Paragraph for Questions 13 & 14

In a $\triangle ABC$ $B(2,4)$, $C(6,4)$ & A lies on a curve S such that $\tan\frac{B}{2}\tan\frac{C}{2} = \frac{1}{2}$.

13. Let a line passing through C and perpendicular to BC intersects the curve S at P and Q. It R is the midpoint of BC then area of $\triangle PQR$ is

(A) $\frac{18}{3}$ sq. units (B) $\frac{8}{3}$ sq. units
(C) $\frac{32}{3}$ sq. units (D) $\frac{26}{3}$ sq. units

14. The eccentricity of the hyperbola, whose transverse axis lies along the line through B, C and the hyperbola passes through B, C & (0, 2)

(A) $\frac{\sqrt{19}}{4}$ (B) $\frac{\sqrt{17}}{2}$
(C) $\sqrt{\frac{7}{3}}$ (D) $\frac{2}{\sqrt{3}}$

Space for rough work

Paragraph for Questions 15 & 16

Let PQRS be a rectangle of size 9×3, if it is folded along QS such that plane PQS is perpendicular to plane QRS & point R moves to point T.

15. Distance between the points P and T will be

(A) $\sqrt{90}$ (B) $\frac{3}{5}\sqrt{205}$

(C) $\frac{4}{5}$ (D) None of these

16. Shortest distance between the edges PQ & TS is

(A) $3\sqrt{\frac{10}{19}}$ (B) $\sqrt{\frac{10}{19}}$

(C) $2\sqrt{\frac{10}{19}}$ (D) None of these

(Match List Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which only **ONE** is correct.

17. Let a, b, c are roots of the equation $x^3 - 3x^2 + 5x - 1 = 0$. A and B are two matrices given as $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ $B = \begin{bmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{bmatrix}$ If α, β, γ are the roots of equation $|B - xI| = 0$ where I is identity matrix of order 3 then match List I with List II:

List I		**List II**	
(P)	The value of det (A) det (B) is	(1)	18
(Q)	The value of $\alpha + \beta + \gamma$ is	(2)	$(18)^2$
(R)	The value of $\|B\|$ is	(3)	$(18)^3$
(S)	The value of $\|A\|$ is	(4)	6
		(5)	$(18)^4$

Codes:

	P	Q	R	S
(A)	4	3	1	2
(B)	3	2	4	1
(C)	3	4	2	1
(D)	4	2	1	3

Space for rough work

18. A straight line passing through O (0, 0) cuts the lines $x = \alpha, y = \beta$ & $x + y = 8$ at A, B, C resp such that OA. OB . OC = $48\sqrt{2}$ & $f(\alpha,\beta) \le 0$ where $f(x,y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{e^x + 2y - e^2 - 6}$.

Match the List -I with List-II:

List-I		**List-II**	
(P)	The value of $2\sqrt{2}(OA + OB + OC)$ is	(1)	50
(Q)	The volume of $OA.OB + OB.OC + O.C.OA$ is	(2)	54
(R)	The value of 25α is	(3)	36
(S)	The value of $18\ \beta$ is	(4)	52

Codes:

	P	Q	R	S
(A)	1	3	4	2
(B)	2	4	1	3
(C)	4	3	1	2
(D)	3	4	1	2

19. Match List I with List II:

List-I		**List-II**	
(P)	If $\alpha = \lim_{y\to\infty} \frac{\int_1^y \left[\tan^{-1} x\right] dx}{\int_1^y \left[1 + \frac{1}{x}\right] dx}$, $\alpha^2 + 3$ is divisible by ([.] denotes greatest integer factor)	(1)	9
(Q)	If $a + b = 3 - \cos 4\theta$ and $a - b = 4\sin 2\theta$, then the maximum value of ab is	(2)	4
(R)	A point D is taken on the side AC of an acute triangle ABC such that AD = 1, DC = 2 and BD is altitude of ΔABC. A circle of radius 2 which passes through points A and D touches the circumcircle of ΔBDC at point D. If area of ΔABC is N then the value of $\frac{N^2}{15}$ is equal to	(3)	1
(S)	$\lim_{x\to\infty} \frac{\sqrt[3]{x^2}\sin(x!)}{(x+1)}$ is less than or equal to	(4)	5
		(5)	0

Codes:

	P	Q	R	S
(A)	3	4	1	1,2,3,4
(B)	1,2	4	1,2,3	1,2
(C)	3	1,4	1,2,3	2,3,4
(D)	1,3	2,4	1,2	2,3

Space for rough work

20. Match List I with List II:

List-I		**List-II**	
(P)	Consider the integral $I_1 = \int_1^e (1+x)(x+\ln x)^{100}\,dx$ $I_2 = \int_{\sin^{-1}1/e}^{\pi/2} (1+e\sin x+\ln\sin x)^{101}\cos x\,dx$ If $I_1 + \frac{e}{101}I_2 = \frac{e(1+e)^{101}-k}{101}$, then K	(1)	3/2
(Q)	$I(k) = \int_0^\infty \frac{\ln x}{x^2+kx+k^2}dx (k>0)$ & $kI(x) - I(1) = \frac{\pi}{\sqrt{3}}, k = e^\alpha$ α is equal to	(2)	1
(R)	Let f(x) be a continuous function satisfying $9f(3) \le \int_0^3 (x^2 f'(x) + (f(x))^2)dx + k$, then the least passible value of k is	(3)	5
(S)	The values of definite integral $\int_1^a x\, a^{-[\log_a x]}dx$ where a > 1 & [.] denotes the greatest integer is $\frac{e-1}{2}$, then the value of 5[a] is	(4)	2
		(5)	9

Codes:

	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	5	3
(C)	2	1	4	3
(D)	2	4	3	1

Space for rough work